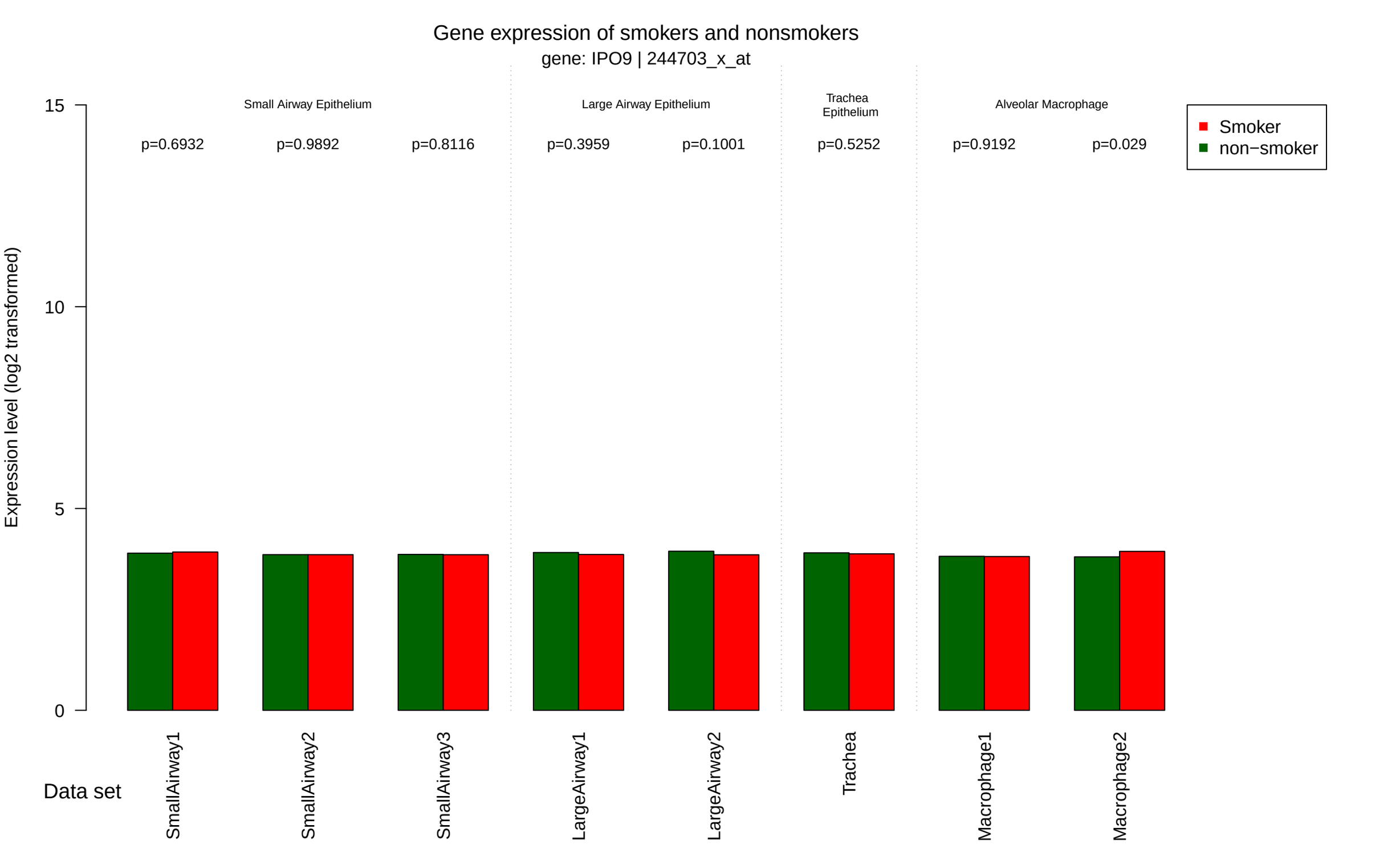
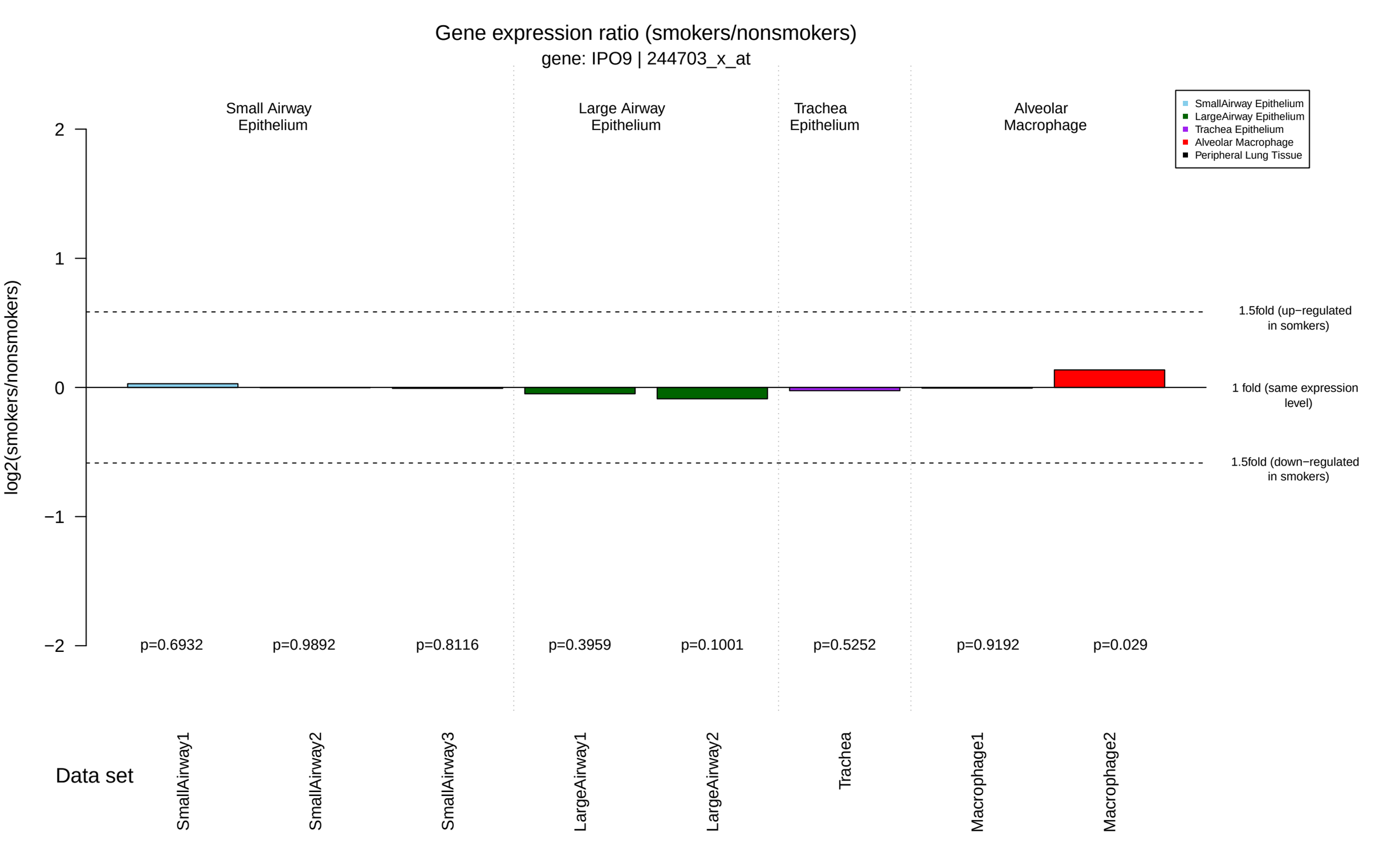
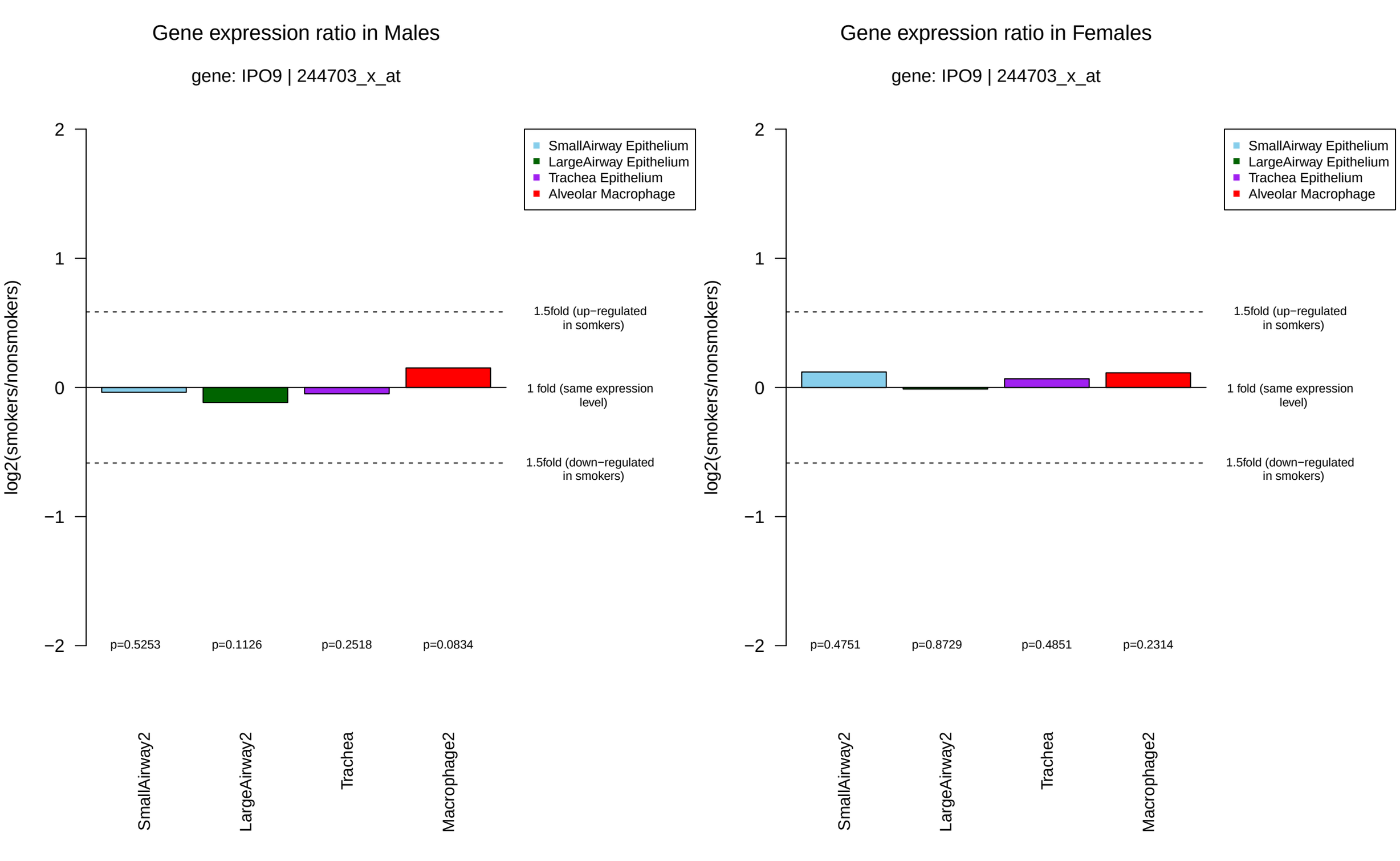
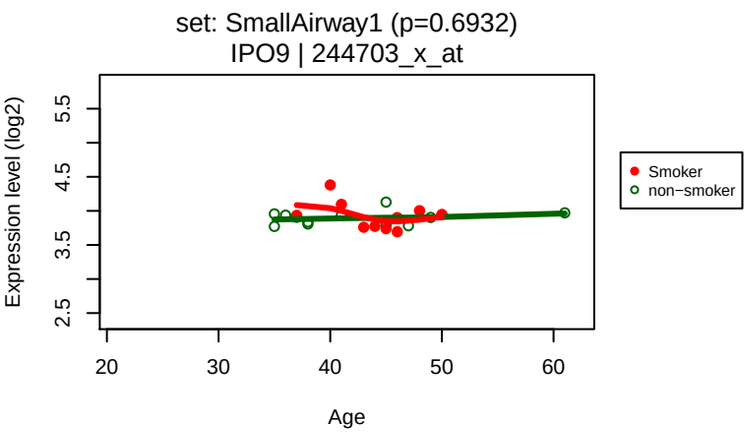
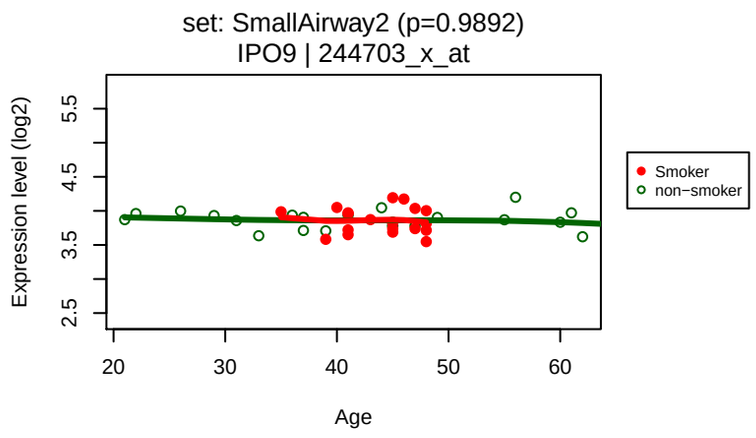
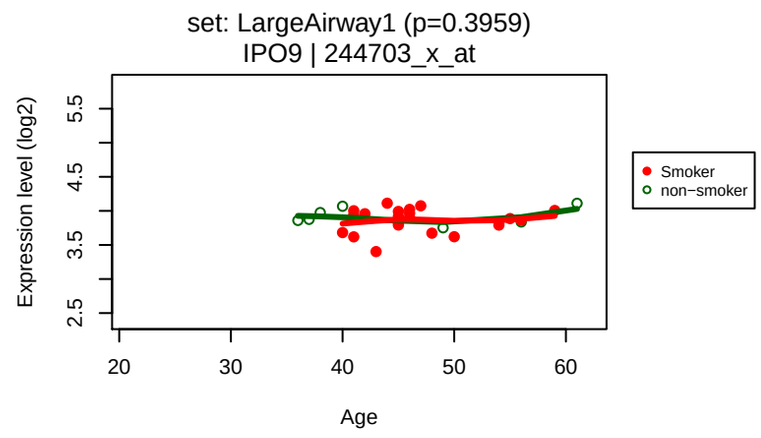

Gene expression of smokers and nonsmokers
gene: IPO9 | 244703_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6932
p=0.9892
p=0.8116
p=0.3959
p=0.1001
p=0.5252
p=0.9192
p=0.029
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

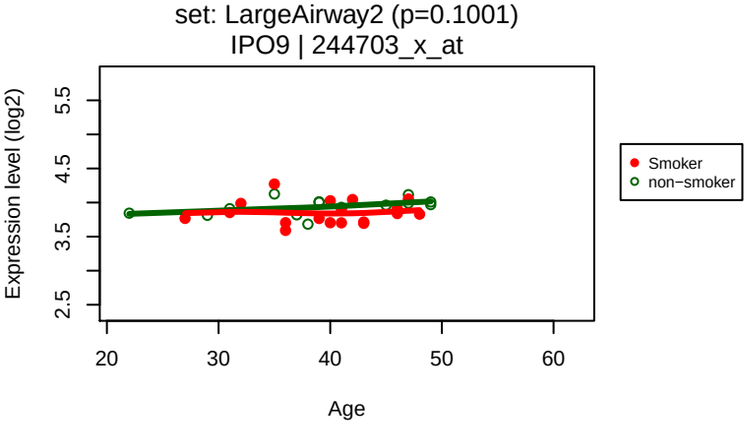

Gene expression ratio (smokers/nonsmokers)
gene: IPO9 | 244703_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6932
p=0.9892
p=0.8116
p=0.3959
p=0.1001
p=0.5252
p=0.9192
p=0.029
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

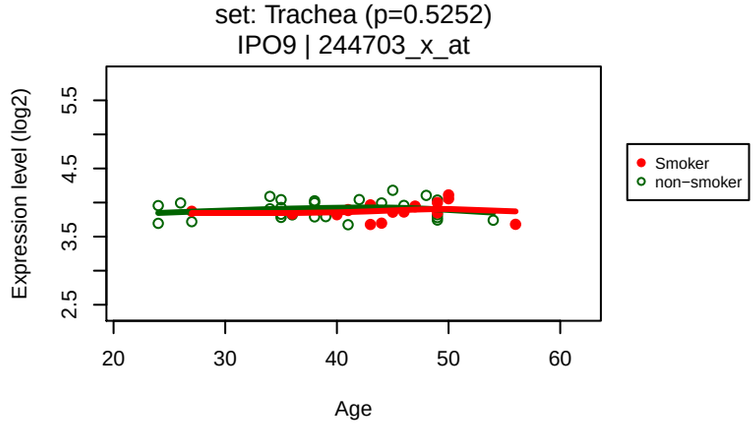

Gene expression ratio in Males
gene: IPO9 | 244703_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5253
p=0.1126
p=0.2518
p=0.0834
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IPO9 | 244703_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4751
p=0.8729
p=0.4851
p=0.2314
SmallAirway2
LargeAirway2
Trachea
Macrophage2

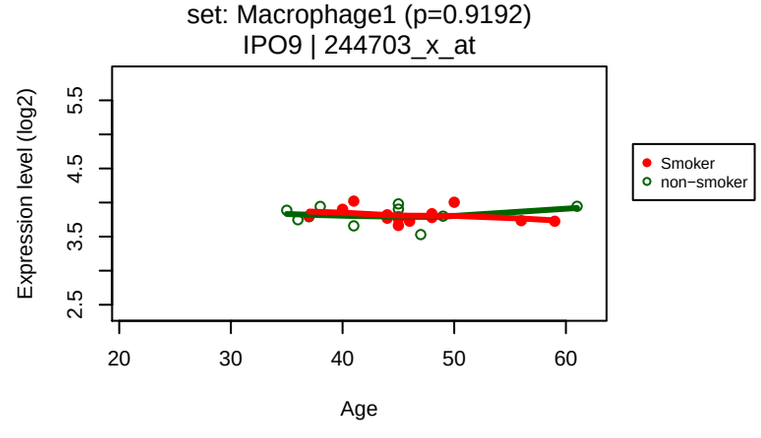

set: SmallAirway1 (p=0.6932)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker

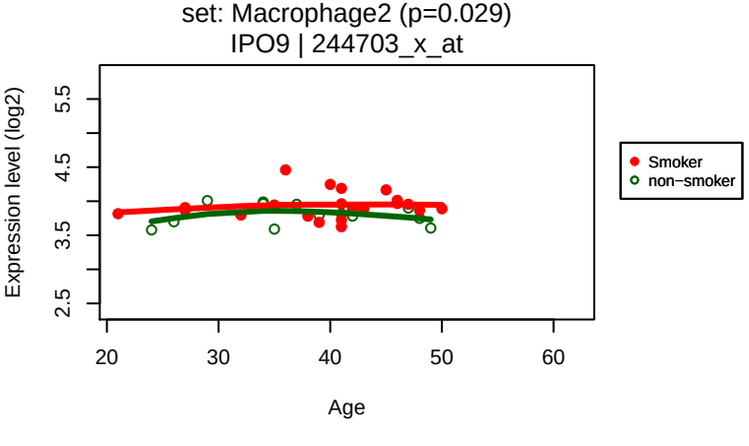

set: SmallAirway2 (p=0.9892)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker

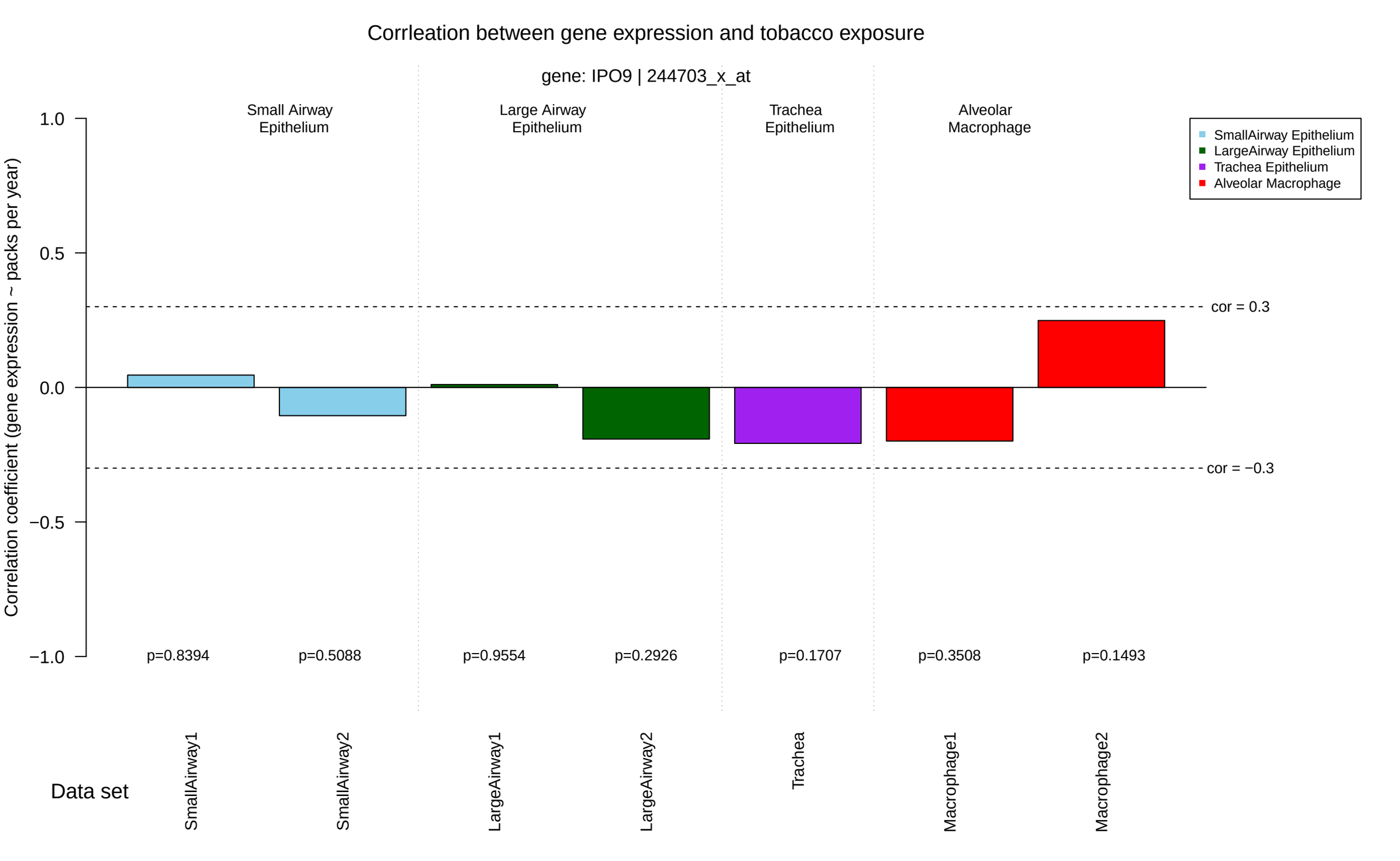

set: LargeAirway1 (p=0.3959)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1001)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5252)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9192)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.029)
IPO9 | 244703_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IPO9 | 244703_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8394
p=0.5088
p=0.9554
p=0.2926
p=0.1707
p=0.3508
p=0.1493
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2